Hyperaware.

A spatial governance protocol
for connected devices.

Our connected world.

We have developed the ability to coordinate on a scale never before possible.

Connected sensors like phones and IoT devices capture, process and share more and more information about their surroundings and users.

Cellular and other wireless networks are increasing in coverage and capacity.

Different countries have different rules for governing vehicles, devices and data.

Missing opportunities.

Device owners have little control over the data produced by their sensors.

Firms aren’t willing to share sensitive information, even if it could benefit others.

Governments use dated ways to apply policies to vehicles and people in their jurisdictions.

These inefficiencies cost companies time and money, and let bad actors break the law.

The Hyperaware Protocol.

Hyperaware is a spatial governance protocol for connected devices, designed for self-sovereign users.

An alpha implementation is running on Ethereum and IoTeX testnets, and on Arweave at https://hyperaware.io.

The protocol defines an (optionally) privacy-preserving and automatic way to apply policies to devices based on their location and other attributes, using enclave computing, smart contracts and spatial analytics algorithms.

Zone Registry.

Authorities register DIDs describing geographic zones and policies to our Zone Registry contract, on Ethereum and Arweave.

Zone owners have complete sovereignty over their Zone borders and policies - including what payments are required and, soon, when a zone policy is in effect, what devices it applies to, who is notified when a device enters or exits, what information is shared, and even whether spatial and policy data is private.

Device Registry.

Owners of location-enabled devices (phones and IoT sensors) register on the Hyperaware Device Registry contract, staking funds to pay Zone fees.

Device owners also control their device identities using a DID, and store the DID document on Arweave.

Additional trust and privacy is possible with trusted IoT devices like IoTeX Pebble tracker.

The protocol is compatible with legacy location tracking devices as well - no new hardware needs to be installed.

Private Location Analytics.

As connected vehicles travel, their positions are streamed into the Hyperaware Private Location Analytics Engine, where we analyze location data inside a secure enclave.

If a vehicle is detected inside a Zone, that zone's policies are automatically enforced by invoking a smart contract method.

Use Cases.

Trace shipments of arms to make sure they aren't diverted and used by terrorist groups.

Track fishing vessels to make sure they have not caught their fish in protected waters.

Follow medicines and be certain they are being delivered to the intended patients in developing countries.

Prevent drones from flying into restricted airspace, causing air accidents or costly delays to travelers.

Track the movements of people infected with Covid-19, making sure they stay quarantined.

All while respecting the user's right to privacy.

Team.

John IV.	Technical Director, London Blockchain Labs. MSc, Spatial Data Science & Visualisation (Distinction), UCL Centre for Advanced Spatial Analysis.
Jason James.	Lead Developer. Founder, iotxplorer. MSc, Computer Science, UCL.
Leo Unoki.	Smart Contract Developer. BSc, Computer Science, Imperial.
Tony Wu.	Front End Developer. BSc, Information Management for Business, UCL.

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■ Next.

Protocol research: white paper. Looking for grant funding and researchers and advisors with expertise in international law and security engineering.

Commercialization. Looking for users, partners, customers, advisors.

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